

A Machine-Checked Proof of the Johnson–Lindenstrauss Lemma in Lean 4

Formalized over mathlib

Abstract

We present a complete formalization, in the Lean 4 proof assistant on top of `mathlib`, of the Johnson–Lindenstrauss (JL) lemma: a random Gaussian projection of n points from $\mathbb{R}^d$ into $\mathbb{R}^m$ with $m = O(\log n/\varepsilon^2)$ dimensions preserves all pairwise squared distances to within a factor $(1 \pm \varepsilon)$, with high probability and *independently of the ambient dimension d* . The development compiles end to end with no `sorry` and depends only on the three standard foundational axioms. We describe the mathematical argument, its decomposition into seven independently verified modules, and the principal Lean techniques used at each step.

1 What we prove, and the plan

The statement. Fix $0 < \varepsilon < 1$ and n distinct points $x_1, \dots, x_n \in \mathbb{R}^d$. Let A be an $m \times d$ random matrix with i.i.d. $N(0, 1/m)$ entries. The Johnson–Lindenstrauss lemma asserts that, provided $m \gtrsim \log n/\varepsilon^2$,

$$(1 - \varepsilon) \|x_i - x_j\|^2 \leq \|Ax_i - Ax_j\|^2 \leq (1 + \varepsilon) \|x_i - x_j\|^2 \quad \text{for all } i \neq j,$$

simultaneously, except with probability at most δ . Concretely, the failure probability is at most δ as soon as $2n^2 \exp(-m\varepsilon^2/12) \leq \delta$, i.e. $m \geq 12(\log(2n^2) + \log(1/\delta))/\varepsilon^2$.

The strategy. We follow the elementary proof of Dasgupta and Gupta [1]. Because A is Gaussian, for a *fixed* vector v the image Av is itself Gaussian, $Av \sim N(0, (\|v\|^2/m)I_m)$, so $\|Av\|^2$ is a scaled chi-squared variable. Concentration of the chi-squared distribution (via its moment generating function and the Chernoff bound) controls the distortion of a single vector; a union bound over the $\binom{n}{2}$ difference vectors $x_i - x_j$ finishes the proof. The argument splits into seven steps, realized as seven Lean modules (Table 1).

Module	Result
LogBounds	§2 elementary log inequalities
GaussianMGF	§3 $\mathbb{E}[e^{tZ^2}] = (1 - 2t)^{-1/2}$, $Z \sim N(0, 1)$
Basic, ChiSq	§4 the law χ_m^2 and its MGF $(1 - 2t)^{-m/2}$
TailBounds	§5 Chernoff tail bounds for χ_m^2
Projection	§6 single-vector concentration
UnionBound	§7 union bound $\Rightarrow$ JL

Table 1: The seven modules and their results.

The formal theorem. The random projection is modelled by its m independent standard-Gaussian rows $G = (G_1, \dots, G_m)$, $G_k \in \mathbb{R}^d$; the projected squared norm of v is $(1/m) \sum_k \langle v, G_k \rangle^2 = \|Av\|^2$ for $A = G/\sqrt{m}$. The law of G is `rowLaw m d`, and $\mu.\text{real } S = (\mu S).\text{toReal}$.

```

theorem johnsonLindenstrauss (n d : N) (m : N) (m_pos : 0 < m)
  (x : Fin n → EuclideanSpace R (Fin d)) (hx : Function.Injective x)
  (eps delta : R) (he_pos : 0 < eps) (he_lt : eps < 1) (hd_pos : 0 < delta)
  (hsuff : 2 * (n : R)^2 * rexp (-(m : R) * eps^2 / 12) <= delta) :
  (rowLaw m d).real
    {G | exists i j, i != j /\
      ((1+eps) * ||x i - x j||^2 <= (1/(m : R)) * sum k, <x i - x j, G k>^2
        \ / (1/(m : R)) * sum k, <x i - x j, G k>^2 <= (1-eps) * ||x i - x j||^2)}
    <= delta

```

2 Logarithm bounds

The Chernoff exponents are $\varepsilon - \log(1 + \varepsilon)$ and $-(\varepsilon + \log(1 - \varepsilon))$. Integrating the pointwise inequality $\frac{1}{1+x} \leq 1 - x + x^2$ over $[0, \varepsilon]$,

$$\varepsilon - \log(1 + \varepsilon) = \int_0^\varepsilon \left(1 - \frac{1}{1+x}\right) dx \geq \int_0^\varepsilon (x - x^2) dx = \frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3},$$

and symmetrically $\varepsilon + \log(1 - \varepsilon) \leq -\varepsilon^2/2$. For $0 < \varepsilon < 1$ both simplify to the clean form used downstream:

$$\varepsilon - \log(1 + \varepsilon) \geq \frac{\varepsilon^2}{6}, \quad \varepsilon + \log(1 - \varepsilon) \leq -\frac{\varepsilon^2}{6}.$$

Remark. The frequently quoted bound $\varepsilon - \log(1 + \varepsilon) \geq \varepsilon^2/3$ is *false* near $\varepsilon = 1$ (at $\varepsilon = 1$ the left side is $1 - \log 2 \approx 0.307 < 1/3$). The sharp constant of this form valid on all of $(0, 1)$ is $\varepsilon^2/6$, since $\frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3} \geq \frac{\varepsilon^2}{6} \iff 2\varepsilon^2(1 - \varepsilon) \geq 0$. We use $\varepsilon^2/6$ throughout; this is one place where formalization corrected a folklore constant.

In Lean the comparison is an interval-integral inequality. The antiderivative $\int_0^\varepsilon \frac{1}{1+x} = \log(1 + \varepsilon)$ comes from a substitution (`integral_comp_add_left`) and `integral_one_div`; the comparison is `intervalIntegral.integral_mono_on`, with integrability discharged by `Continuous.intervalIntegrable` and the pointwise step by `nlinarith`.

3 The MGF of a squared standard normal

For $Z \sim N(0, 1)$ and $t < 1/2$, completing the square inside the Gaussian integral,

$$\mathbb{E}[e^{tZ^2}] = (2\pi)^{-1/2} \int e^{tx^2} e^{-x^2/2} dx = (2\pi)^{-1/2} \int e^{-(\frac{1}{2}-t)x^2} dx = (2\pi)^{-1/2} \sqrt{\frac{\pi}{\frac{1}{2}-t}} = (1 - 2t)^{-1/2}.$$

The Lean proof passes to the density via `integral_gaussianReal.eq_integral_smul`, applies mathlib's Gaussian integral `integral_gaussian` ($\int e^{-bx^2} = \sqrt{\pi/b}$), and discharges the radical identity by normalizing both sides to $\sqrt{(1 - 2t)^{-1}}$ (`Real.sqrt_inv`, `Real.sqrt_mul`, `field.simp`, `Real.sqrt_eq_rpow`).

4 The chi-squared distribution and its MGF

The chi-squared law with m degrees of freedom is the pushforward of the standard Gaussian on $\mathbb{R}^m$ under $x \mapsto \|x\|^2$:

```
noncomputable def chiSq (m : N) : Measure R :=
  (stdGaussian (EuclideanSpace R (Fin m))).map (fun x => ||x||^2)
```

Writing the standard Gaussian on $\mathbb{R}^m$ as the pushforward of $\bigotimes N(0, 1)$ makes $\|g\|^2 = \sum_i g_i^2$, so the MGF *factorizes*:

$$\mathbb{E}[e^{t\|g\|^2}] = \prod_{i=1}^m \mathbb{E}[e^{tg_i^2}] = ((1 - 2t)^{-1/2})^m = (1 - 2t)^{-m/2}.$$

The key mathlib ingredient is `integral_fintype_prod_eq_prod`, the product-measure factorization $\int \prod_i f_i(x_i) d(\bigotimes \mu) = \prod_i \int f_i$; the m -fold product is collapsed with `rpow_natCast` and `rpow_mul`.

Lemma 1 (`chiSq_mgf`). *For $t < 1/2$, $\text{mgf id } (\chi_m^2) t = (1 - 2t)^{-m/2}$.*

5 Chernoff tail bounds

Applying the Chernoff method to `chiSq_mgf` and optimizing t ($t = \varepsilon/(2(1 + \varepsilon))$ for the upper tail, $t = -\varepsilon/(2(1 - \varepsilon))$ for the lower),

$$\mathbb{P}(\chi_m^2 \geq (1 + \varepsilon)m) \leq e^{-\frac{m}{2}(\varepsilon - \log(1 + \varepsilon))} \leq e^{-m\varepsilon^2/12}, \quad \mathbb{P}(\chi_m^2 \leq (1 - \varepsilon)m) \leq e^{\frac{m}{2}(\varepsilon + \log(1 - \varepsilon))} \leq e^{-m\varepsilon^2/12}.$$

The Lean tail bounds invoke `measure_ge_le_exp_mul_mgf` and its lower counterpart; integrability of e^{tX} is free, since the MGF is finite and nonzero so the integrand cannot be non-integrable (`integral_undef`, contrapositive). The exponent identity reduces, after `Real.rpow_def_of_pos`, to `field.simp`; `ring`, and the simplification to $e^{-m\varepsilon^2/12}$ chains through §2.

6 Single-vector concentration

For fixed $v \neq 0$, $\|Av\|^2 \sim (\|v\|^2/m)\chi_m^2$. We model this law as `projNormSq m v = ( $\chi_m^2$ ).map( $z \mapsto (\|v\|^2/m)z$ )` and reduce each distortion event to a chi-squared tail by change of variables (`Measure.map_apply`), obtaining

$$\mathbb{P}(\|Av\|^2 \notin [(1 - \varepsilon)\|v\|^2, (1 + \varepsilon)\|v\|^2]) \leq 2e^{-m\varepsilon^2/12}.$$

The two-sided event is split with `measureReal_union_le` and the two §5 bounds applied via `gcongr`.

7 The union bound and the JL lemma

Two ingredients complete the proof.

The marginal law (the technical crux). With m i.i.d. standard-Gaussian rows, the projected norm of a fixed v has law exactly `projNormSq m v`:

```
lemma rowLaw_map_projSq (m d : N) (v : EuclideanSpace R (Fin d)) :
  (rowLaw m d).map (fun G => (1/(m:R)) * sum i, <v, G i>^2) = projNormSq m v
```

This is established by a chain of pushforwards: each coordinate $\langle v, G_k \rangle \sim N(0, \|v\|^2)$ via `IsGaussian.map_eq_gauss` (mean 0, variance $\|v\|^2$ from `variance_dual_stdGaussian` and `innerSL_apply_norm`); these lift coordinatewise through `Measure.pi_map_pi`; a rescaling (`gaussianReal_map_const_mul`) and the identity $\sum (\|v\| y_i)^2 = \|v\|^2 \sum y_i^2$ collapse the result onto `chiSq`, reusing `map_pi_eq_stdGaussian`.

The union bound. The bad event is contained in the union, over the $\leq n^2$ distinct pairs, of the per-pair distortion events. By sub-additivity (`measureReal_biUnion_finset_le`) and the per-pair bound (§6 via the marginal),

$$\mathbb{P}(\text{bad}) \leq \sum_{i \neq j} 2e^{-m\varepsilon^2/12} \leq n^2 \cdot 2e^{-m\varepsilon^2/12} = 2n^2 e^{-m\varepsilon^2/12} \leq \delta.$$

Injectivity of the points gives $x_i - x_j \neq 0$ (`sub.ne_zero`), as the per-pair bound requires. This yields the theorem of §1.

8 Assessment

Trust. `#print axioms johnsonLindenstrauss` reports exactly [`propext`, `Classical.choice`, `Quot.sound`] — the standard `mathlib` foundation, with no `sorryAx`. Every intermediate lemma, including the marginal, was checked the same way.

The bound. The explicit sufficient condition $2n^2 e^{-m\varepsilon^2/12} \leq \delta$ gives $m = O(\log n/\varepsilon^2)$, independent of d — the entire point of the lemma. The constant 12 is honest, tracing through the $\varepsilon^2/6$ logarithm bound and the factor $1/2$ in the chi-squared exponent.

Engineering. The proof is seven small files (≈ 500 lines), each compiled independently and mirroring one step of the argument. Small, named lemmas were preferred to monolithic tactic blocks at every turn: they are easier to state, to prove, to read, and to repair.

References

- [1] S. Dasgupta and A. Gupta. *An elementary proof of a theorem of Johnson and Lindenstrauss*. Random Structures & Algorithms 22(1):60–65, 2003.
- [2] W. B. Johnson and J. Lindenstrauss. *Extensions of Lipschitz mappings into a Hilbert space*. Contemp. Math. 26:189–206, 1984.
- [3] The mathlib Community. *The Lean mathematical library*. CPP 2020.